6 Digit Nixie Clock "64Bit" - Driver Board

64Bit comes from the fact that this clock is driven with shift registers. Between the 6 nixie digits, and indicators, 64 bits are sent from the microcontroller to the shift registers to create the display output

This is an STM32F446 based project.

The clock time keeping is handled via a Real Time Clock (RTC) communicated with via I2C. The STM32 also has an on-board RTC that can be utilized. Both RTCs have a battery backup via a replaceable 2032 coin cell.

The interface with the clock is via a rotory encoder with a push button. Interfacing with remote capacitive buttons is possible via connectors on the bottom of the Driver Board.

There is an ESP32-C6 module driving a full-color LCD screen. The ESP32 features wifi and bluetooth allowing for automatic time setting and smart home integration. Communication with the STM32 is via UART.

This clock was designed with safety in mind. There is a dedicated safety circuit that cuts the input and output connections of the high voltage power supply via relays. There are connectors for interfacing external spring pin connections to the cover of the clock, ensuring the high voltage power supply is not on while the lid is off. The safety circuit features dual channels and feedback to the microcontroller. With the spring pin feature, it will provide isolation from the high voltage, even in an event where the microcontroller experiences an error.

The power design of the clock relys on simple components. There are two inputs available to power the clock. A USB-C port and a 12V barrel jack. A boost converter utilizing XL6009 converts (boost) the incoming voltage to 12VDC. Then an LDO creates 5V then another LDO, in series, creates 3.3V. The high voltage power supply typically is set to output 170VDC to 180VDC. It is supplied by a power supply module that sources 12V power.

This driver board interfacese with the display board PCB via right angle board-to-board connectors. Mounted to the back of the display board is an LED board to provide RGB backlighting to each IN-12 Nixie Tubes.

The circuit boards will be mounted to a 3D printed base via brass standoffs. Overtop of the PCBs will be a glass top. The glass top will have sections where copper tape is applied. Spring pins contact the copper tape to pass connections for the safety circuit.

STM32 MCU

I2C

Interface

Safety Circuit

Power

Connectors

Connectors 2

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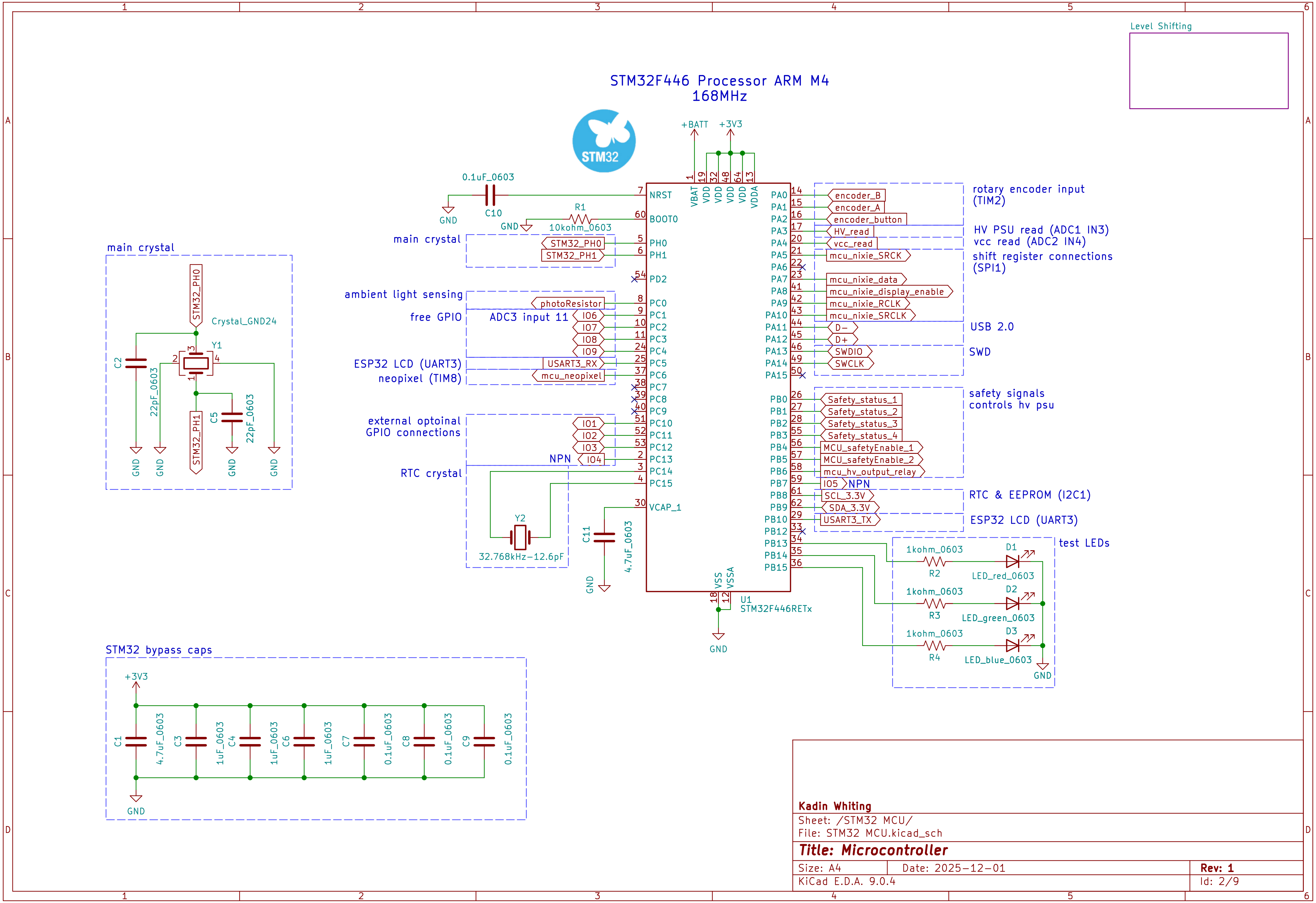
Sheet: /
File: 64 Bit - Main Board.kicad_sch

Title: Project Overview

Size: A4	Date: 2025-12-01	Rev: 1
KiCad E.D.A. 9.0.4		Id: 1/9



STM32F446 Processor ARM M4
168MHz
Level Shifting
main crystal
STM32_PH0
STM32_PH1
Crystal_GND24
Y1
C2
22pF_0603
C5
22pF_0603
GND
0.1uF_0603
C10
GND
R1
10kohm_0603
+BATT
+3V3
VBAT
VDD
VDDA
NRST
BOOT0
PH0
PH1
PD2
ambient light sensing
photoResistor
ADC3 input 11
free GPIO
IO6
IO7
IO8
IO9
ESP32 LCD (UART3)
USART3_RX
neopixel (TIM8)
mcu_neopixel
external optoinal
GPIO connections
IO1
IO2
IO3
IO4
NPN
RTC crystal
Y2
32.768kHz-12.6pF
VCAP_1
C11
4.7uF_0603
VSS
VSSA
U1
STM32F446RETx
rotary encoder input
(TIM2)
encoder_B
encoder_A
encoder_button
HV PSU read (ADC1 IN3)
vcc read (ADC2 IN4)
HV_read
vcc_read
shift register connections
(SPI1)
mcu_nixie_SRCK
mcu_nixie_data
mcu_nixie_display_enable
mcu_nixie_RCLK
mcu_nixie_SRCLK
USB 2.0
D-
D+
SWD
SWDIO
SWCLK
safety signals
controls hv psu
Safety_status_1
Safety_status_2
Safety_status_3
Safety_status_4
MCU_safetyEnable_1
MCU_safetyEnable_2
mcu_hv_output_relay
IO5
NPN
RTC & EEPROM (I2C1)
SCL_3.3V
SDA_3.3V
ESP32 LCD (UART3)
USART3_TX
test LEDs
1kohm_0603
R2
D1
LED_red_0603
R3
D2
LED_green_0603
R4
D3
LED_blue_0603
STM32 bypass caps
C1
4.7uF_0603
C3
1uF_0603
C4
1uF_0603
C6
1uF_0603
C7
0.1uF_0603
C8
0.1uF_0603
C9
0.1uF_0603
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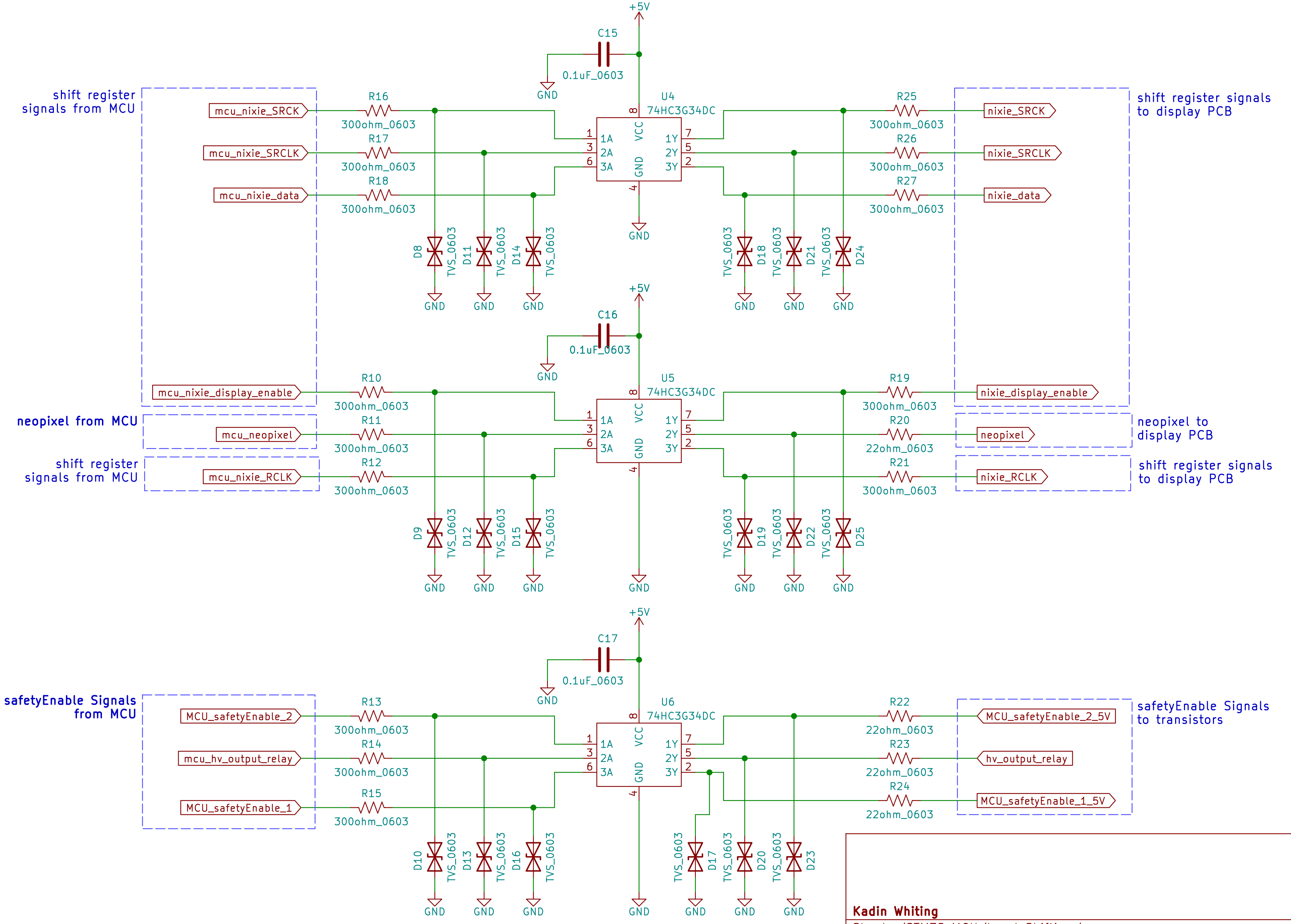
Sheet: /STM32 MCU/
File: STM32 MCU.kicad_sch
Title: Microcontroller
Size: A4
Date: 2025-12-01
Rev: 1
KiCad E.D.A. 9.0.4
Id: 2/9



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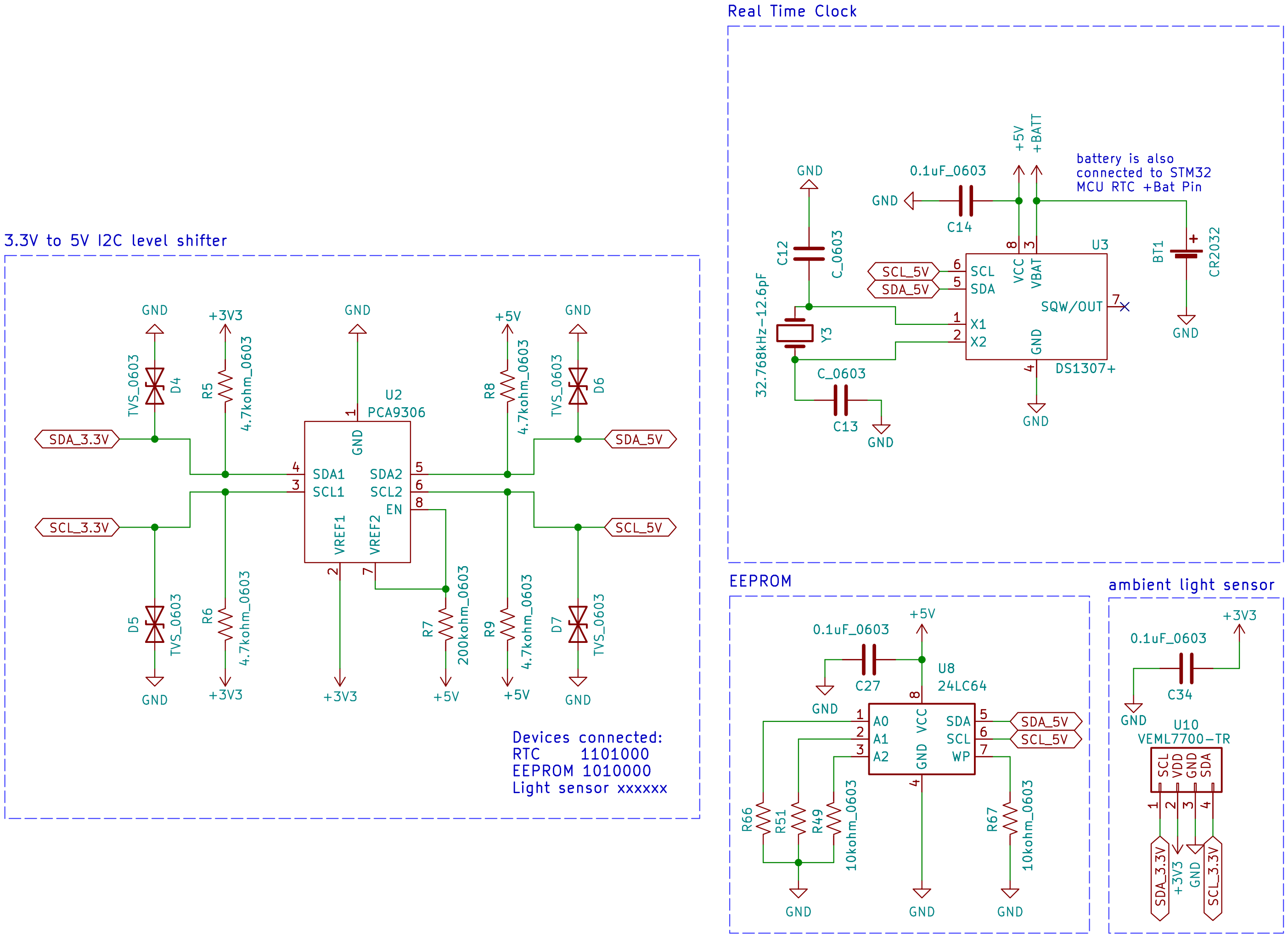
Sheet: /STM32 MCU/Level Shifting/

File: Level Shifting.kicad_sch

Title: 3.3V -> 5.0V Level Shifting

Size: A4 | Date: 2025-12-01 | **Rev: 1**

KiCad E.D.A. 9.0.4 | Id: 4/9



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Sheet: /I2C/

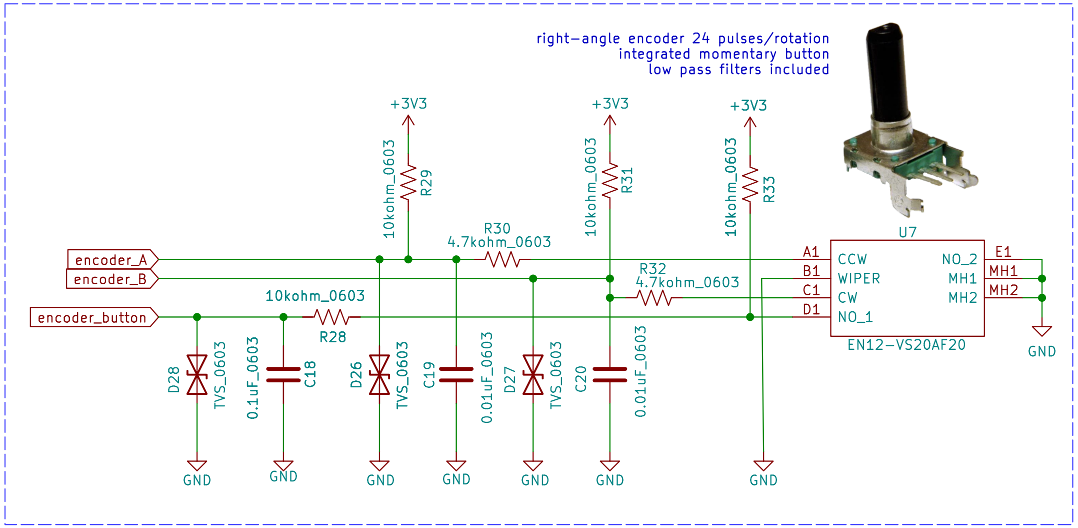
File: I2C.kicad_sch

Title: RTC & RTC Level Shifting

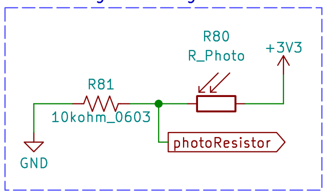
Size: A4 | Date: 2025-12-01 | **Rev: 1**

KiCad E.D.A. 9.0.4 | Id: 3/9

rotary encoder with push button



photoresistor
ambient light sensing



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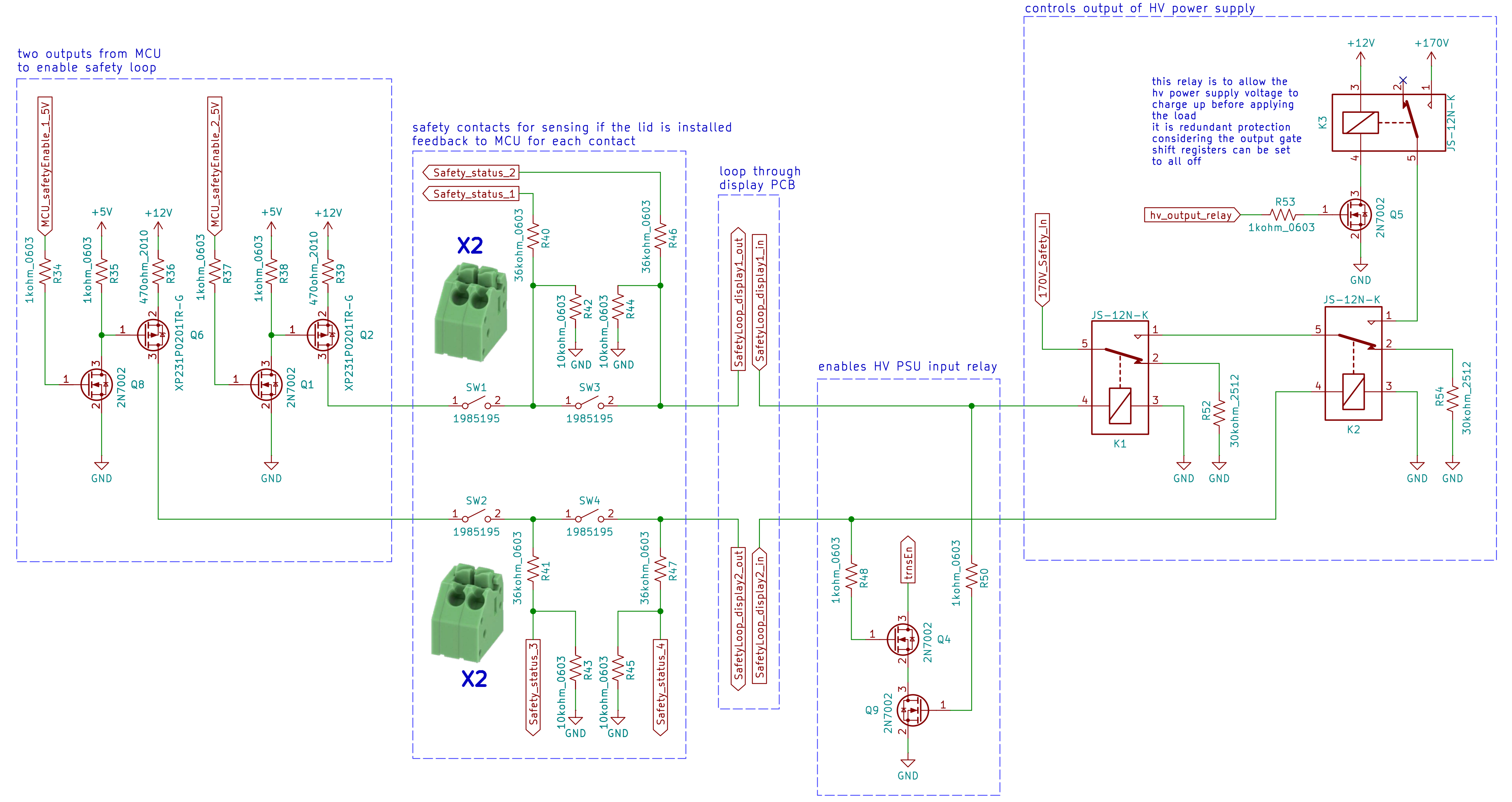
Sheet: /Interface/
File: Interface.kicad_sch

Title: Interface – Rotary Encoder

Size: A4	Date: 2025-12-01	**Rev: 1**
KiCad E.D.A. 9.0.4		Id: 5/9



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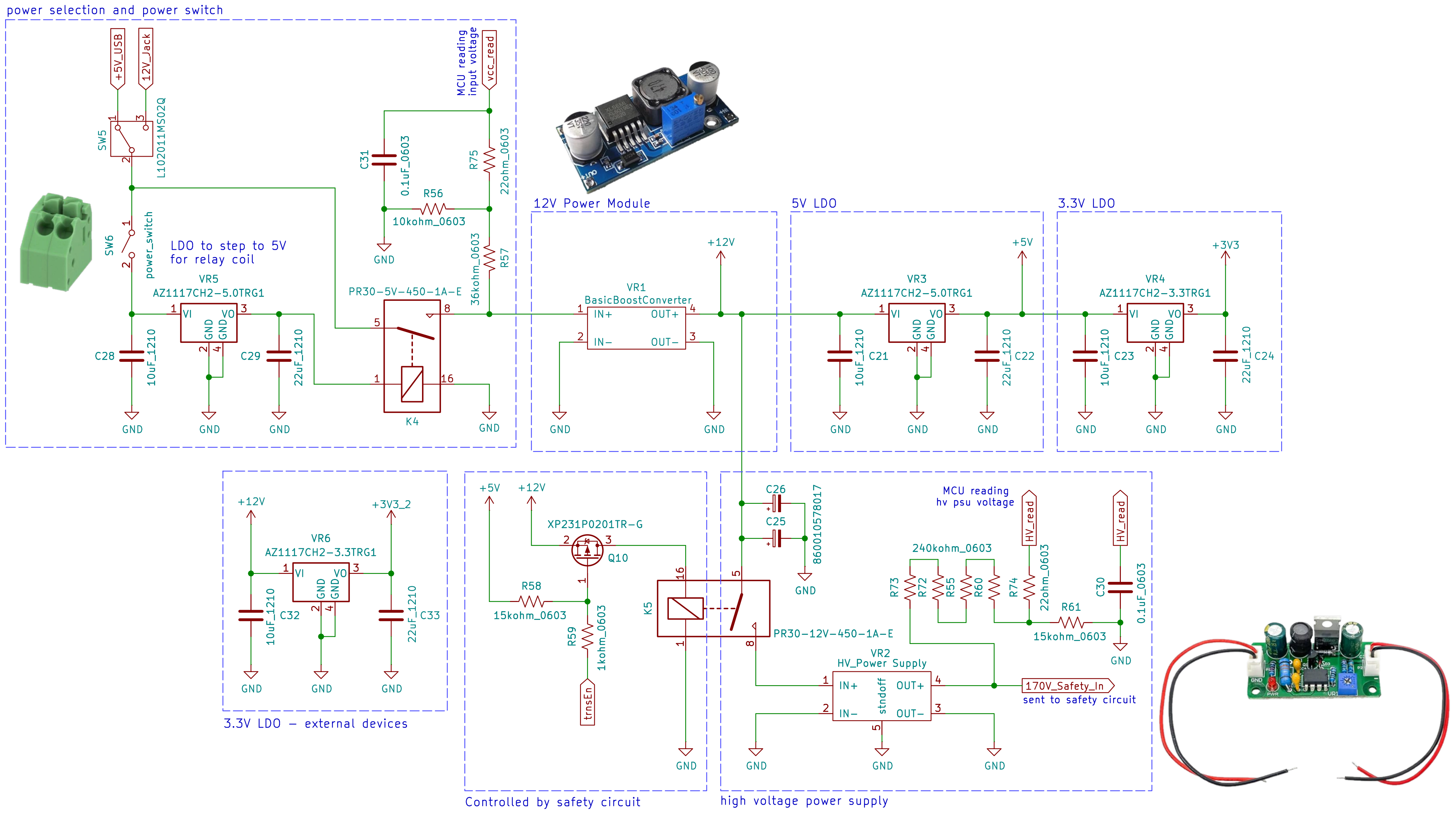
Sheet: /Safety Circuit/
File: Safety Circuit.kicad_sch

Title: Safety Circuit — High Voltage PSU Control

Size: A4	Date: 2025-12-01	**Rev: 1**
KiCad E.D.A. 9.0.4		Id: 6/9



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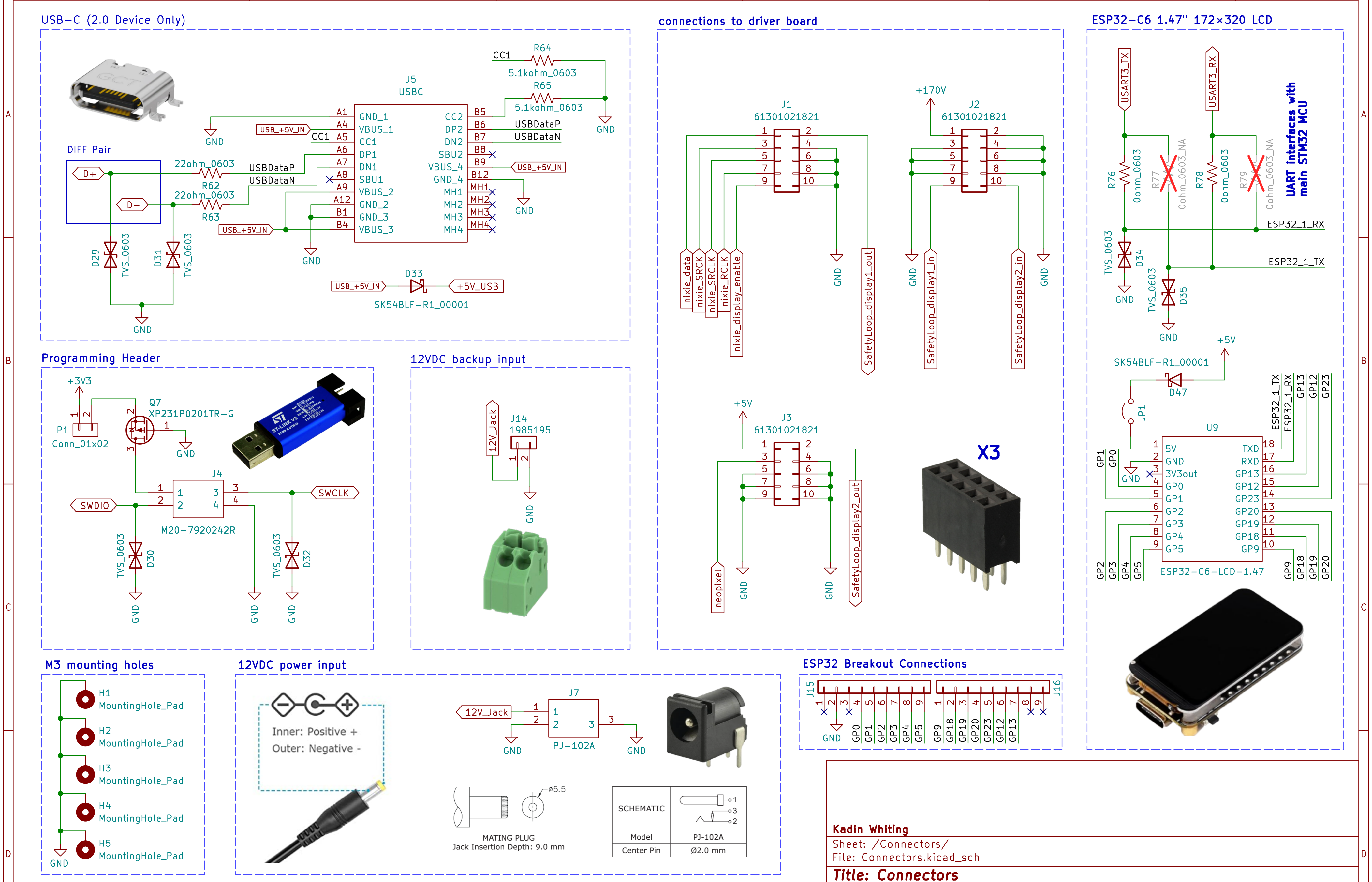
Sheet: /Power/

File: Power.kicad_sch

Title: Power

Size: A4	Date: 2025-12-01	Rev: 1
KiCad E.D.A. 9.0.4		Id: 7/9



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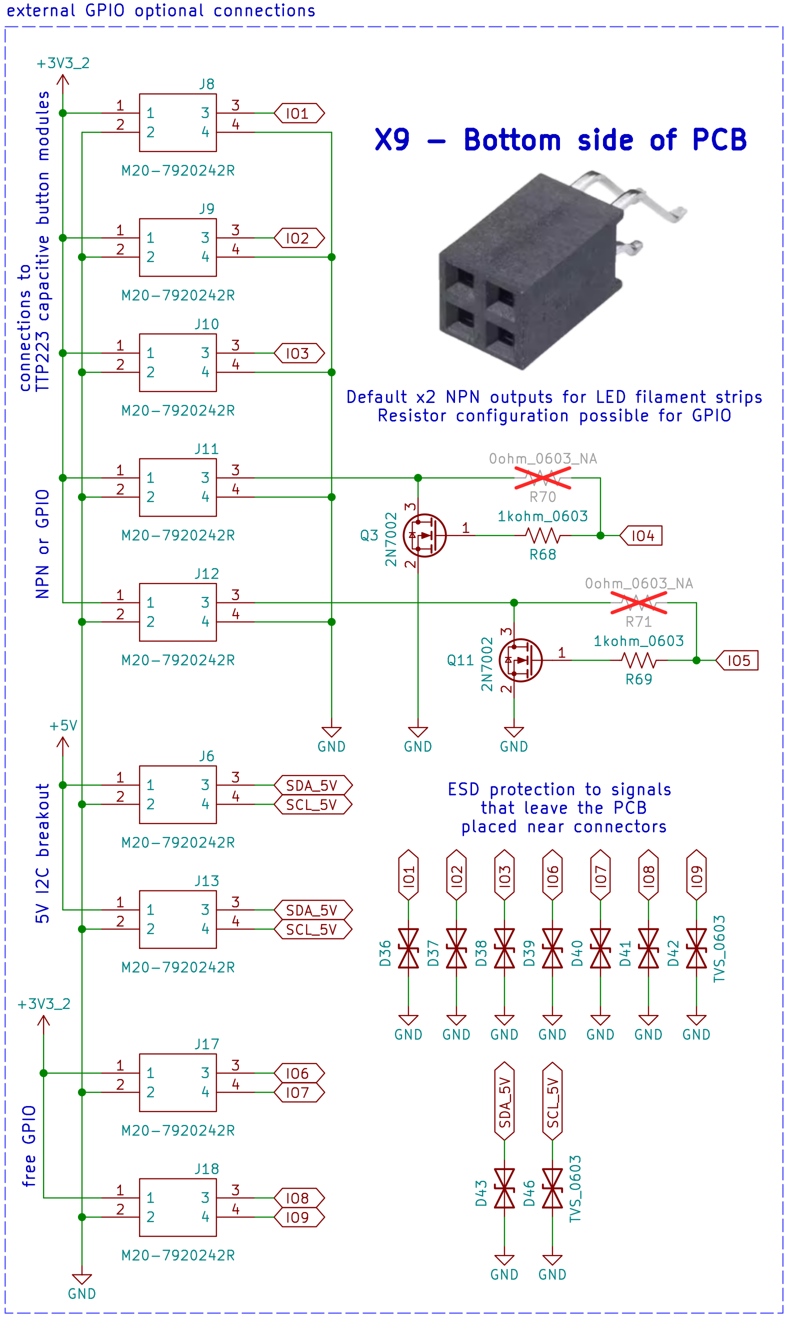
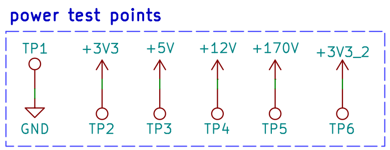
Sheet: /Connectors/

File: Connectors.kicad_sch

Title: Connectors

Size: A4	Date: 2025−12−01	Rev: 1
KiCad E.D.A. 9.0.4		Id: 8/9





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Sheet: /Connectors 2/
File: Connectors 2.kicad_sch

Title: Connectors

Size: A4	Date: 2025-12-01	**Rev: 1**
KiCad E.D.A. 9.0.4		Id: 9/9